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EDUCATION

Cairo University Giza, Egypt
BSc in Computer Science — Computer Science Major. GPA: 3.30 (Distinction) May 2027

- Top 30 on Dean's List. Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Object-Oriented Programming.

EXPERIENCE

Route Academy — Backend .NET Course (In Progress) Giza, Egypt
Backend Trainee Feb 2026 – Jul 2026

- Comprehensive backend course covering C#, ASP.NET Core, Entity Framework Core, REST APIs, JWT Authentication, LINQ, SQL Server, and software engineering principles including SOLID, Design Patterns, and Clean Architecture.
- Built Uni-Researcher-System: a complete database from SRS to real structure, executing complex high-performance queries and applying database administration roles.

DEPI — AI Track, Microsoft Egypt Initiative (Hybrid) Egypt
AI Trainee Jun 2025 – Dec 2025

- Government-backed tech initiative covering foundational AI concepts and Microsoft Azure cloud basics (AZ-900 level), with hands-on cloud deployment exposure.

Coach Academy — Competitive Programming Remote
Competitive Programming Trainee May 2024 – Oct 2024

- Solved 700+ Algorithms & Data Structures problems in C++, building strong analytical and problem-solving foundations.

Udemy — CS Basics Course (Remote) Remote
Student — Instructed by Dr. Mostafa Saad (Nvidia, PhD) Sep 2023 – Apr 2025

- Applied foundational CS concepts through projects including sparse data structures and the Vole Machine simulator.

PROJECTS

Contact-Manager CLI — Enterprise Contact Management System | C#, OOP, SOLID Principles, JSON Storage, Clean Architecture

- Developed a production-grade contact management system demonstrating enterprise software architecture patterns and clean code principles aligned with industry best practices.
- Implemented comprehensive CRUD operations with advanced filtering and search capabilities, JSON-based persistent storage, and a user-friendly CLI interface supporting high-volume contact operations.
- Applied SOLID principles (Single Responsibility, Open/Closed, Liskov Substitution, Interface Segregation, Dependency Inversion) to create a scalable, maintainable, and extensible codebase architecture.
- Designed the system to replace legacy phone-book-style lookup systems with a modern, modular architecture enabling seamless feature expansion and integration.

Course Management System — Educational Platform & Database Design | C#, Database Design, ERD Modeling, Business Logic, Data Management

- Built an end-to-end course and student management platform managing complex relationships between instructors, students, and course catalogs with role-based access controls.
- Designed and implemented a normalized relational database schema with Entity-Relationship Diagrams (ERD), ensuring data integrity, optimal query performance, and scalable data operations.
- Engineered business logic for course enrollment, instructor course assignments, student progress tracking, and administrative operations demonstrating understanding of enterprise business workflows.
- Applied database best practices including transaction management, constraint enforcement, and query optimization to support institutional-scale academic operations.

Your Eyes — AI-Powered Assistive Vision System | Python, Machine Learning, Computer Vision, Innovation, Real-World Problem Solving

- Developed an intelligent computer vision system leveraging deep learning to assist visually impaired individuals through glasses-integrated camera technology, demonstrating commitment to social impact and digital innovation.
- Implemented end-to-end machine learning pipeline from data collection and preprocessing through model training, optimization, and real-time inference for assistive applications.
- Engineered object detection and scene understanding algorithms enabling contextual information delivery, showcasing expertise in cutting-edge AI technologies and their practical applications in digital transformation.
- Demonstrated ability to translate emerging technologies (ML/AI) into user-centric solutions addressing real societal challenges, aligning with enterprise innovation and CSR initiatives.

Tharwa Project — Investment & Financial Management Platform | HTML, JavaScript, Full-Stack Development, Software Engineering Lifecycle, Team Collaboration

- Designed and developed a complete investment management application from requirements gathering through deployment, applying the full software development lifecycle and agile methodologies in a team environment.
- Built a comprehensive financial platform enabling users to manage diverse asset portfolios (stocks, real estate, commodities), perform investment analysis, and track portfolio performance with real-time data aggregation.
- Implemented secure user authentication, multi-asset portfolio management, and advanced financial calculations demonstrating proficiency in building user-centric business applications.
- Collaborated with cross-functional team members throughout the project lifecycle, demonstrating communication, problem-solving, and adaptability skills essential for enterprise development environments.

Uni-Research-System — University Database | SQL Server, Database Administration

- Designed full database structure from SRS to physical schema, executed complex business-requirement queries, and applied database administration for access control and role management.

SKILLS & INTERESTS

Programming Languages: Python (Proficient), C++ (Fluent), C# (Intermediate), Java, SQL

Frameworks & Libraries: .NET Core, ASP.NET Core, Entity Framework Core, REST API, LINQ, JWT Authentication, Postman, Microsoft SQL Server

Concepts: Design Patterns, MVC Architecture, System Architecture, SOLID Principles, Object-Oriented Programming (OOP)

Cloud: Microsoft Azure — AZ-900 (basic exposure)

Soft Skills: Communication, Leadership, Design Thinking, Critical Thinking, Problem-Solving; 2× Hackathon Winner; 10+ public talks & presentations

Languages: Arabic (Native), English (Full Professional Proficiency — C1)

Interests: Competitive programming, teaching & mentoring, content creation (16+ educational YouTube videos)

RELEVANT ACTIVITIES / VOLUNTEERING

Problem Setter — Internal Programming Contest Feb 2026 – Mar 2026

- Crafted problems from ideation to final judging across 4 rounds; used Polygon to verify optimal solutions and calibrate difficulty to participants' level. Honoured by the Dean of the Faculty.

Instructor — Fatema Elfiky Educational YouTube Channel Nov 2024 – Present

- Published 16+ videos explaining challenging CS topics to fellow students, reflecting a commitment to collaborative learning and knowledge sharing.

WINNER (2nd Place) — Finology Competition, Central Bank of Egypt Jun 2025
Hackathon

- Built Smart Budgeters: an interactive fintech tool promoting financial literacy among youth through smart budgeting, expense tracking, and money-management features.

ECPC Qualifications, Alexandria — Contestant Aug 2024 & Aug 2025

- Ranked 123rd / 400 teams (Aug 2024) and improved to 80th / 270 teams (Aug 2025), demonstrating consistent growth in competitive programming performance.